Goal: Be able to understand the notation $\sum_{i=1}^{N} a_i$ and approximate the (signed) area beneath a curve.

Sigma (Summation) Notation

We want shorthand for sums with a predictable pattern. We use the Greek letter Sigma (Σ) to do so.

$$\sum_{i=m}^{n} a_i \overset{\text{def}}{=} a_m + a_{m+1} + \cdots + a_n$$

"Variable" (index) → $i=m$ ← where to start; a_i ← i^{th} term

Example 1. Compute $\sum_{i=1}^{4} i^2$

$$\sum_{i=1}^{4} i^2 = 1^2 + 2^2 + 3^2 + 4^2 = 1 + 4 + 9 + 16 = 30$$

Example 2. Compute $\sum_{j=0}^{6} (2^j + 1)$

$$\sum_{j=0}^{6} (2^j+1) = (2^0+1)+(2^1+1)+(2^2+1)+(2^3+1)+(2^4+1)+(2^5+1)+(2^6+1)$$
$$= 2 + 3 + 5 + 9 + 17 + 33 + 65 = 134$$

Example 3. Rewrite $2^3 + 3^3 + 4^3 + \ldots + n^3$ in sigma notation.

i^3 when $i = n$

$$\sum_{i=2}^{n} i^3 = 2^3 + 3^3 + \cdots + n^3$$

Remark (Re-indexing)**.** You can start from any integer if you adjust your formula accordingly

$$\sum_{i=2}^{n} i^3 = \sum_{j=1}^{n-1} (j+1)^3 = \sum_{j=0}^{n-2} (j+2)^3 = \sum_{j=-1}^{n-3} (j+3)^3$$

Replace i with $j+1$

Example 4. Express $1^2 + 2^2 + (\sqrt{2}+1)^2 + (\sqrt{3}+1)^2 + \ldots + (\sqrt{n}+1)^2$ using sigma notation.

$(\sqrt{0}+1)^2$ $(\sqrt{1}+1)^2$ k

$$\sum_{k=0}^{n} (\sqrt{k}+1)^2$$

Theorem. Sums are linear in the sense that if k is a constant (i.e., not a function of the index), then:

(i) $\displaystyle\sum_{i=m}^{n} (a_i \pm b_i) = \sum_{i=m}^{n} a_i \pm \sum_{i=m}^{n} b_i$

(ii) $\displaystyle\sum_{i=m}^{n} k a_i = k \sum_{i=m}^{n} a_i$

Example 5. Compute $\displaystyle\sum_{i=1}^{n} k$ for any constant k.

$$\sum_{i=1}^{n} 1 = \underbrace{1+1+1+\cdots+1}_{n \text{ times}} = n$$

$$\sum_{i=1}^{n} k = k \sum_{i=1}^{n} 1 = kn$$

Example 6. Show $\sum_{i=1}^{n} i = 1 + 2 + 3 + \ldots + n = \dfrac{n(n+1)}{2}$

$S = 1 + 2 + 3 + \cdots + (n-1) + n$

$+ \; S = n + (n-1) + (n-2) + \cdots + 2 + 1$

$2S = \underbrace{(n+1) + (n+1) + \cdots + (n+1)}_{n \text{ times}}$

$2S = n(n+1)$

$S = \dfrac{n(n+1)}{2}$

The next theorem is not required, but it is beneficial to know since it can speed up calculations.

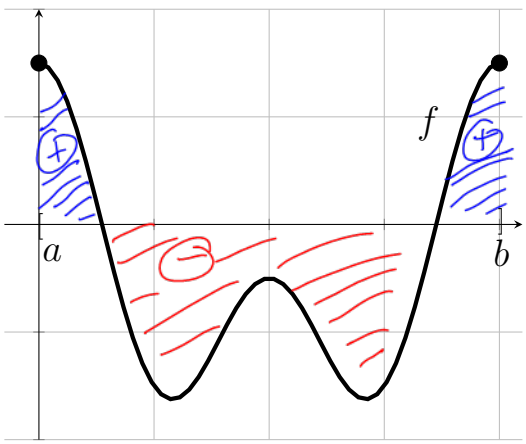
Theorem. (i) $\sum_{i=1}^{n} k = kn$ for constant k

(ii) $\sum_{i=1}^{n} i = \dfrac{n(n+1)}{2}$

(iii) $\sum_{i=1}^{n} i^2 = \dfrac{n(n+1)(2n+1)}{6}$

(iv) $\sum_{i=1}^{n} i^3 = \left[\dfrac{n(n+1)}{2}\right]^2$

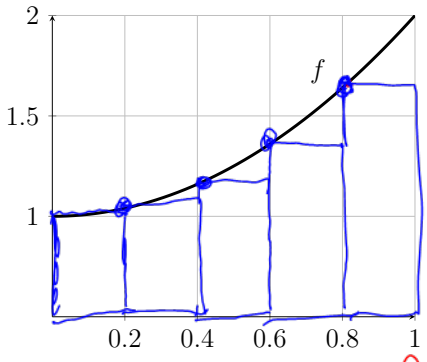
The Area Problem



Problem: Given a function f defined on a closed interval $[a, b]$, compute the (signed) area between the graph of f and the x-axis.

"Solution:" Approximate the region using rectangles.

Example 7. Approximate the area of the region beneath the graph of $f(x) = x^2 + 1$ on $[0, 1]$ using 5 rectangles.



$$\text{Width} = \frac{\text{Length of } [0,1]}{5} = \frac{1-0}{5} = \frac{1}{5}$$

$$\text{Area} \approx \sum \text{Area of Rectangle}$$

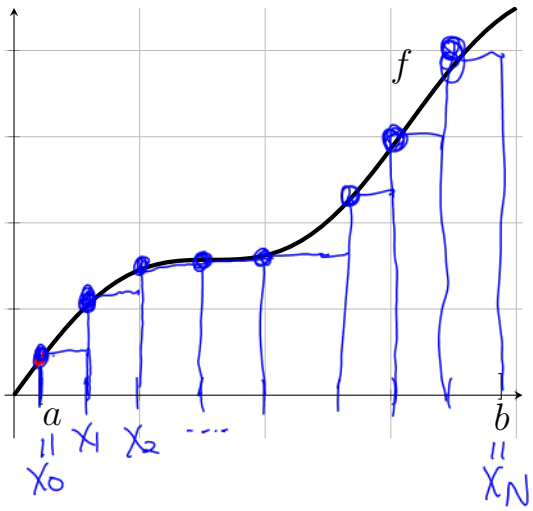
$$= \frac{1}{5} f(0) + \frac{1}{5} f\left(\frac{1}{5}\right) + \frac{1}{5} f\left(\frac{2}{5}\right) + \frac{1}{5} f\left(\frac{3}{5}\right) + \frac{1}{5} f\left(\frac{4}{5}\right)$$

$$= \sum_{i=0}^{4} \frac{1}{5} \cdot f\left(\frac{i}{5}\right) = \sum_{i=0}^{4} \frac{1}{5} \left[\left(\frac{i}{5}\right)^2 + 1\right]$$

$$= \frac{1}{5} \sum_{i=0}^{4} \left[\left(\frac{i}{5}\right)^2 + 1\right] = \frac{1}{5}\left[(0^2+1) + \left(\frac{1}{5}\right)^2 + 1 + \left(\frac{2}{5}\right)^2 + 1 + \left(\frac{3}{5}\right)^2 + 1 + \left(\frac{4}{5}\right)^2 + 1\right]$$

$$= \frac{1}{5}\left[5 + \frac{6}{5}\right] = 1 + \frac{6}{25} = 1.24$$

This is an example of a **left** Riemann Sum ("left" is referring to how we chose the height). How can we do this in general?



1. Divide $[a, b]$ into N equal regions.
2. Determine the dimensions of each rectangle.

Width: $\frac{\text{Length of } [a,b]}{N} = \frac{b-a}{N} \overset{\text{def}}{=} \Delta x$

Height: $x_i = a + i\Delta x$ for $0 \le i \le N-1$

Height: $f(x_i) = f(a + i\Delta x)$

Definition. The **left Riemann sum** of f over $[a, b]$ is defined as: (using N rectangles)

$$L_N \overset{\text{def}}{=} \sum_{i=0}^{N-1} f(a + i\Delta x)\,\Delta x$$

($f(a+i\Delta x)$: Height; Δx: width)

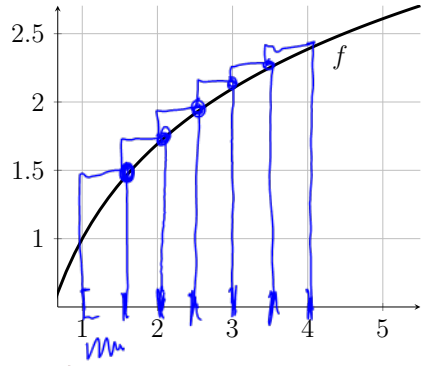
Similarly, the **right Riemann sum** is defined as:

$$R_N \overset{\text{def}}{=} \sum_{i=1}^{N} f(a + i\Delta x)\,\Delta x$$

where $\Delta x = \frac{b-a}{N}$

Example 8. Consider $f(x) = \ln x + 1$ on the interval $[1, 4]$:

(a) Compute R_6 [i.e., the right Riemann sum with 6 rectangles].



Width: $\Delta x = \frac{b-a}{N} \overset{\substack{a=1\\ b=4\\ N=6}}{\Longrightarrow} \frac{4-1}{6} = \frac{3}{6} = \frac{1}{2}$

$$R_6 = \sum_{i=1}^{6} f\left(1 + i\left(\tfrac{1}{2}\right)\right)\cdot\tfrac{1}{2} = \tfrac{1}{2}\sum_{i=1}^{6} f\left(1 + \tfrac{i}{2}\right)$$

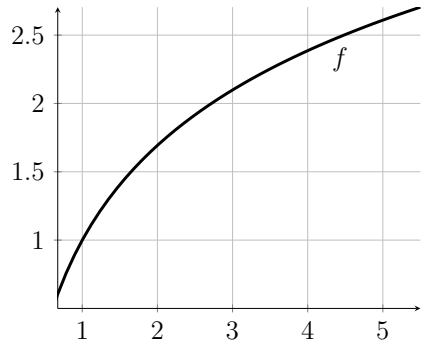
$$= \tfrac{1}{2}\sum_{i=1}^{6}\left[\ln\left(1+\tfrac{i}{2}\right)+1\right] = \tfrac{1}{2}\sum_{i=1}^{6} 1 + \tfrac{1}{2}\sum_{i=1}^{6}\ln\left(1+\tfrac{i}{2}\right)$$

$$= 3 + \tfrac{1}{2}\left[\ln\left(\tfrac{3}{2}\right)+\ln(2)+\ln\left(\tfrac{5}{2}\right)+\ln(3)+\ln\left(\tfrac{7}{2}\right)+\ln(4)\right]$$

$$= 3 + \tfrac{1}{2}\left[\ln\left(\tfrac{3}{2}\cdot 2\cdot\tfrac{5}{2}\cdot 3\cdot\tfrac{7}{2}\cdot 4\right)\right]$$

$$\approx 5.876286$$

(b) Repeat for L_6



(c) Set up (but do not compute) L_N and R_N.

Width:

$$\Delta X = \frac{b-a}{N} \xrightarrow[b=4]{a=1} \Delta X = \frac{4-1}{N} = \frac{3}{N}$$

$$L_N = \sum_{i=0}^{N-1} f(a+i\Delta X)\Delta X = \sum_{i=0}^{N-1} \left[\ln\left(1+i\left(\frac{3}{N}\right)\right)+1\right]\frac{3}{N}$$

$$R_N = \sum_{i=1}^{N} \left[\ln\left(1+\frac{3i}{N}\right)+1\right]\cdot\frac{3}{N}$$